1

i

Our goal is to show that $T^* = \arg\max_{T \in X} \Pi_{e \in T}(1 - p_e) \equiv \arg\min_{T \in X} \sum_{e \in T} p_e$

We can follow a similar proof shown in class:

$$T^* = \arg\max_{T \in X} \Pi_{e \in T}(1 - p_e)$$
$$\equiv \arg\max_{T \in X} \ln(\Pi_{e \in T}(1 - p_e))$$
$$= \arg\max_{T \in X} \sum_{e \in T} \ln(1 - p_e)$$
$$\equiv \arg\min_{T \in X} - \sum_{e \in T} \ln(1 - p_e) \equiv T^*$$

Let T' denote $\arg\min_{T \in X} \sum_{e \in T} p_e$ or the MST of a graph whose edge weights are $w_e = p_e$

Suppose $T' \neq T^*$ then $\exists e_1 \in T' \wedge e_1 \notin T^*$ and $\exists e_2 \notin T' \wedge e_2 \in T^*$

From T' and from the Optimal Message Passing Problem we determine $0 < p_{e1} < p_{e2} < 1$

Thus it is clear that $1 - p_{e1} > 1 - p_{e2}$

From T^* we determine that $-\ln(1 - p_{e2}) < -\ln(1 - p_{e1}) \to \ln(1 - p_{e2}) > \ln(1 - p_{e1})$

This is a contradiction because the ln function is strictly increasing function meaning if $x_1 > x_2 \to \ln x_1 > \ln x_2$ which is not the case shown above.

Thus $T^* = T'$ because $T^* \neq T'$ creates a contradiction.

ii

Claim: $\exists R : (X, p) \to (X, w)$ which creates an instance of MST Problem from an instance of Optimal Message Passing Problem by defining weights of the given graph for Optimal Message Passing Problem as follows:

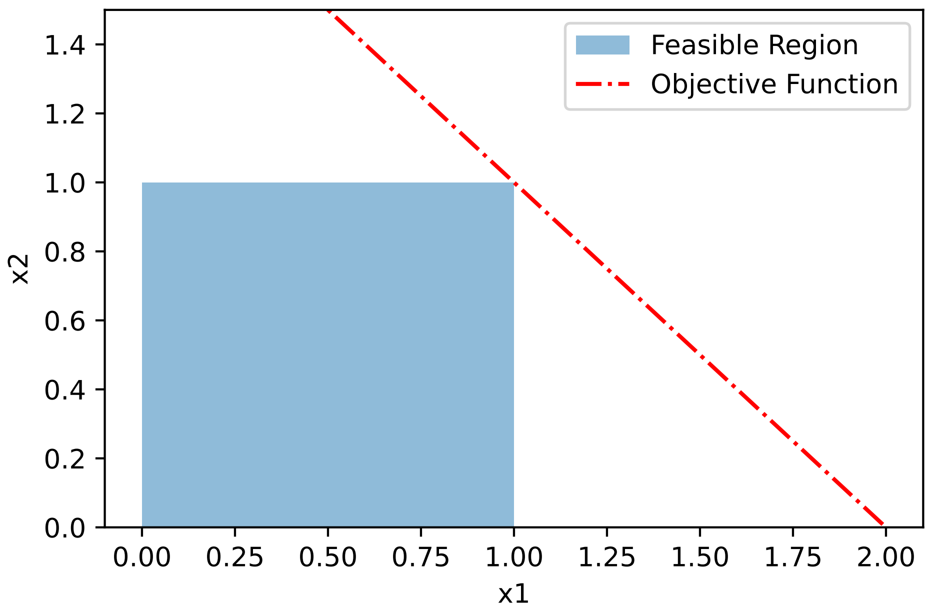
$$\forall e \in E, w^*(e) = p(e)$$

We are claiming that the optimal spanning tree for the MST is also an optimal spanning tree for the Optimal Mssage Passing Problem.

Proof: Given from the Optimal Message Passing Problem that $0 < p(e) < 1$, thus $1 - p(e) > 0$, we are able to apply the transformations done in **1i**. Using **1i**, we come to the conclusion that $T^* = T'$ thus the proposed reduction algorithm R optimally solves Optimal Message Passing Problem using reduction to Minimum Spanning Tree Problem.

2

a



This program has a unique optimal solution which is $x^* = (x_1 = 1, x_2 = 1)$ because the highest value of the objective function is 2 and only intersects at a single point of the bounds $(x_1 = 1, x_2 = 1)$ as shown in the above image.

b

We are given: $\max x_1 + x_2$

s.t. $2x_1 + x_2 \leq 3$

$0 \leq x_1, x_2 \leq 1$

Thus the standard form is:

$\min -y_1 - y_2$

s.t. $2(y_1) + (y_2) + s_1 = 3$

$y_1 + z_1 = 1$

$y_2 + z_2 = 1$

$y_1, z_1, y_2, z_2, s_1 \geq 0$

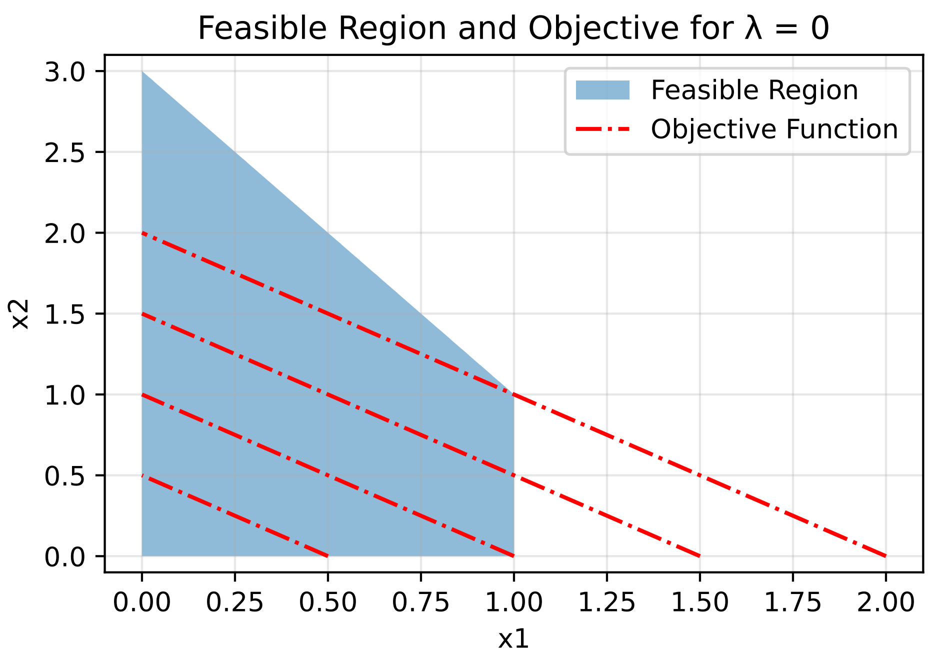
c

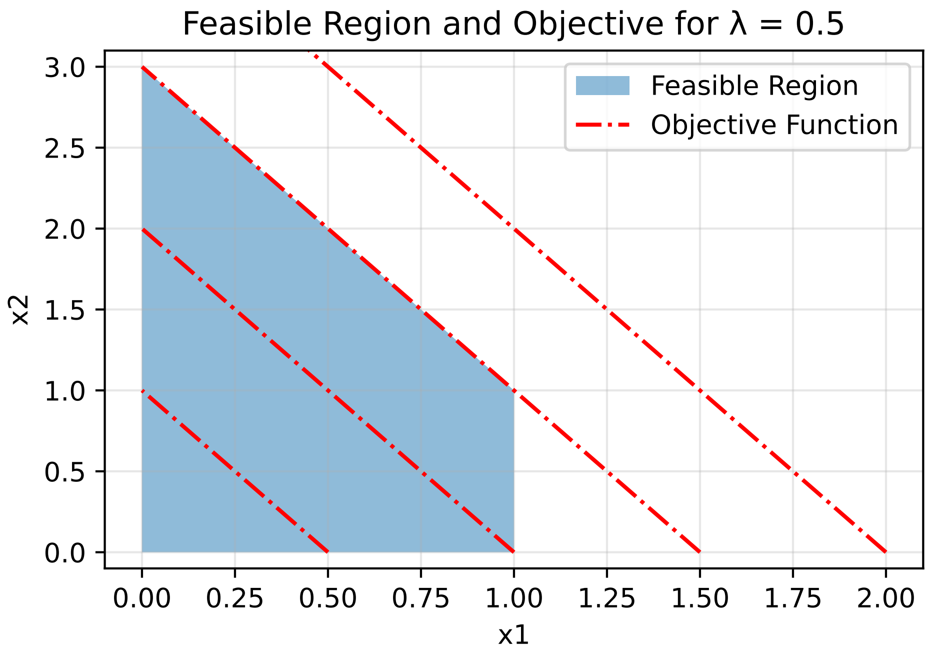
```
Maximize
  obj: x1 + x2
Subject To
  c1: 2 x1 + x2 <= 3
Bounds
  0 <= x1 <= 1
  0 <= x2 <= 1
End
```

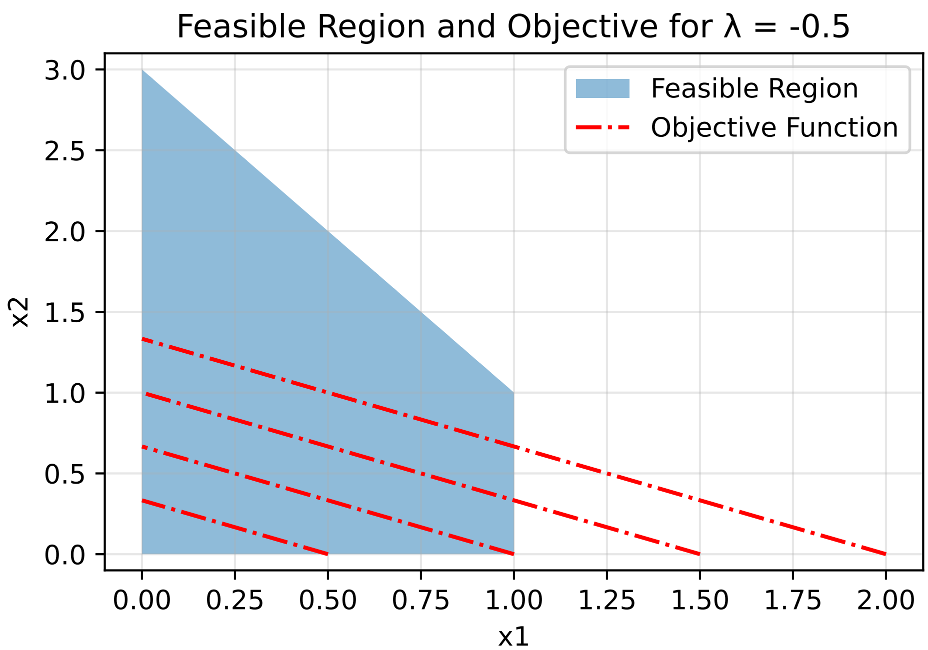
This results in an optimum of $x_1 = 1, x_2 = 1$

3

a







b

We are given: $\max x_1 + (1 - \lambda)x_2$

s.t. $2x_1 + x_2 \leq 3$

$0 \leq x_1 \leq 1$

$0 \leq x_2$

Thus the standard form is:

$\min -y_1 - (1-\lambda)y_2$

s.t. $2(y_1) + (y_2) + s_1 = 3$

$y_1 + z_1 = 1$

$y_1, z_1, y_2, s_1 \geq 0$

c

$\lambda = \frac{3}{4}$ results in the same optimum of $x_1 = 1, x_2 = 1$ as in **2.c**

d